Slide 1 · Domain Overview & TAE 1 Recall

Short-Term Rental & Group Travel Management Database

Problem Statement

- Manages short-term property rentals, guest bookings, and group travel itineraries.
- Transforms raw, unnormalized Airbnb-style listing data into a structured relational model.
- Goal: prevent double-booking, track host revenue, and manage guest reviews reliably.

Final 3NF Relational Schema (approved in TAE 1)

- HOST (Host_ID PK) -- owns -- PROPERTY (Host_ID FK)
- PROPERTY (Property_ID PK) -- has -- BOOKING (Property_ID FK)
- GUEST (Guest_ID PK) -- makes -- BOOKING (Guest_ID FK)
- BOOKING (Booking_ID PK) -- generates -- REVIEW (Booking_ID FK, UNIQUE)
- Constraints: Check_Out > Check_In, Rating 1-5, no overlapping bookings per property

Slide 2 · System Architecture & Implementation Setup

Phase II Implementation Setup

Target RDBMS

MySQL 8.0

InnoDB storage engine, chosen for row-level locking and ACID-compliant transactions -- essential for preventing double-booking during high-concurrency booking periods.

Tools Used

- MySQL Workbench / CLI
- Python 3 + Faker (synthetic data generation)
- GitHub (version control & submission)

Database Topology

- Single-node relational database, 5 core tables
- Referential integrity enforced via FK constraints with ON DELETE / ON UPDATE CASCADE
- 1 trigger enforcing the no-overlap booking rule
- 1 stored procedure for the booking transaction
- 2 reporting views + composite indexes on high-traffic join columns